

TABLE 1: GVI APPLIED TO DESIGN PARAMETERS TO OBTAIN PARAMETER WEIGHT CHANGE

| Customer Requirement   | Change Probability | Rating | Operating Voltage | Max. Current | Control Voltage | NEMA Std | User Interface | Connectivity | Size | After-Sales Service | Weight | Cost | Safety Margin | Parts Env. Norm | Labour Norm | Sustainability Norm | Current Sense | Voltage Sense | Temp. Sense | Humidity Sense | Data Storage Tech | Data Storage Size |
|------------------------|--------------------|--------|-------------------|--------------|-----------------|----------|----------------|--------------|------|---------------------|--------|------|---------------|-----------------|-------------|---------------------|---------------|---------------|-------------|----------------|-------------------|-------------------|
| <b>Performance</b>     |                    |        |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Power rating           | Medium             | 6      | 1                 | 1            |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Service life           | Low                | 9      |                   | 1            |                 | 1        |                |              |      | 1                   |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Warranty               | Low                | 9      |                   |              |                 |          |                |              |      | 1                   |        |      | 1             |                 |             |                     |               |               |             |                |                   |                   |
| Installation           | Low                | 9      |                   |              |                 |          |                |              |      | 1                   |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Maintenance            | Low                | 9      |                   |              |                 |          |                |              |      | 1                   |        |      | 1             |                 |             |                     |               |               |             |                |                   |                   |
| Safety features        | Medium             | 6      |                   |              |                 | 1        |                |              |      |                     |        |      |               |                 |             |                     | 1             | 1             |             |                |                   |                   |
| <b>Physical</b>        |                    |        |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Size                   | Medium             | 6      |                   | 1            |                 | 1        |                |              | 1    |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Weight                 | Medium             | 6      |                   |              |                 |          |                |              |      |                     | 1      |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Environment protection | Medium             | 6      |                   |              |                 | 1        |                |              |      |                     |        | 1    |               |                 |             |                     |               |               |             |                |                   |                   |
| <b>Regulatory</b>      |                    |        |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Environmental hazards  | Medium             | 6      |                   |              |                 |          |                |              |      |                     |        |      |               | 1               |             |                     |               |               |             |                |                   |                   |
| Disposal               | Medium             | 6      |                   |              |                 |          |                |              |      |                     |        |      |               | 1               |             |                     |               |               |             |                |                   |                   |
| Safety norms           | None               | 1      |                   |              |                 | 1        |                |              |      |                     |        |      | 1             | 1               |             |                     |               |               |             |                |                   |                   |
| Labour practice        | None               | 1      |                   |              |                 |          |                |              |      |                     |        |      |               |                 | 1           |                     |               |               |             |                |                   |                   |
| <b>Logistic</b>        |                    |        |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Packaging              | Medium             | 6      |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             | 1                   |               |               |             |                |                   |                   |
| Package dimension      | Low                | 9      |                   |              |                 |          |                |              | 1    |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Nett weight            | Low                | 9      |                   |              |                 |          |                |              |      |                     | 1      |      |               |                 |             |                     |               |               |             |                |                   |                   |
| <b>Economic</b>        |                    |        |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Price                  | High               | 9      |                   |              | 1               |          |                |              |      |                     |        | 1    |               |                 |             |                     |               |               |             |                |                   |                   |
| Profit margin          | High               | 9      |                   |              |                 |          |                |              |      |                     |        | 1    |               |                 |             |                     |               |               |             |                |                   |                   |
| Sales volume           | Medium             | 6      |                   |              |                 |          | 1              |              |      |                     |        | 1    | 1             |                 |             |                     |               |               |             | 1              | 1                 |                   |
| Payback period         | Medium             | 6      |                   |              |                 |          |                |              |      |                     |        | 1    |               |                 |             |                     |               |               |             |                |                   |                   |
| <b>Smart Features</b>  |                    |        |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Power generation       | High               | 9      |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     | 1             | 1             |             |                |                   |                   |
| Power consumption      | High               | 9      |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     | 1             | 1             |             |                |                   |                   |
| User interface         | High               | 9      |                   |              |                 |          | 1              |              |      |                     |        |      |               |                 |             |                     |               |               |             | 1              | 1                 |                   |
| Connectivity           | Low                | 9      |                   |              |                 |          |                | 1            |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   |                   |
| Information storage    | Low                | 9      |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               |               |             |                |                   | 1                 |
| Weather data           | Low                | 9      |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     |               | 1             | 1           |                |                   | 1                 |
| Battery status         | High               | 9      |                   |              |                 |          |                |              |      |                     |        |      |               |                 |             |                     | 1             | 1             |             |                |                   |                   |
| Parameter Change Wt    |                    |        | 6                 | 21           | 9               | 28       | 15             | 9            | 15   | 36                  | 15     | 36   | 25            | 13              | 1           | 6                   | 33            | 33            | 9           | 9              | 15                | 33                |

helping you make informed decisions about when and how to adjust your product's direction. Let us apply the concept of GVI to find out how our charge controller design will be impacted in the future.

## GVI steps

**Change estimation.** The calculation of the GVI relies on insights from our earlier discussion about QFD analysis in the previous articles of our series. In the QFD assessment, we compiled a list of customer expectations. When computing the GVI, we predict the potential change in these expectations over time. This prediction employs a four-point

scale to indicate:

- High (Rating 9)
- Medium (Rating 6)
- Low (Rating 3)
- No change expected (Rating 1)

Once estimated, these values are applied to the design parameters, and their cumulative values create the parameter change weight (PCW) shown in Table 1.

**GVI estimation.** The parameter change weight (PCW) offers a glimpse into which design parameters might evolve in the future. Our next stride involves the second aspect of QFD, the design parameter house. In this stage, we integrate the PCW values and gauge the ease

or difficulty of altering the design components associated with each parameter. Here, the same four-point scale as the PCW estimation is employed:

- High (Rating 9) – Demands significant product redesign
- Medium (Rating 6) – Requires partial component redesign
- Low (Rating 3) – Necessitates simple component alterations
- None (Rating 1) – Entails minimal design changes

Following the evaluation of change difficulty, we multiply the difficulty weight by the PCW and consolidate the values column-wise. For a comprehensive breakdown